CURRENT ELECTRICITY

1. Two identical cells are connected in (a) series(b) parallel then maximum power transfered to the load is :-

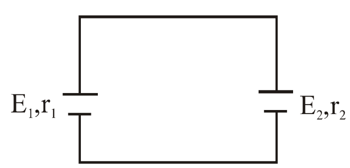
(1) Same in both cases

(2) Different in both cases

(3) May or may not remain same depending upon polarity of the cell connected

(4) None

2. Two cells of unequal emf E_1 and E_2 having internal resistance r_1 and r_2 are connected as shown in figure :-



(1) Potential difference across any of the cell cannot be zero

(2) Potential difference across cell of smaller EMF may be zero

(3) Potential difference across cell of larger EMF may be zero

(4) Potential difference can be zero only if EMF of both cell are same

3. Conductivity of electrolyte is very low as compared to metal at room temperature because :-

(1) Charge of each ion of electrolyte is very large as compare to charge of each free e^-

(2) Free ion density in electrolyte is much smaller as compared to e^- density of metal

(3) Free ion density in electrolyte is very large as compared to e^- density of metal

(4) None

4. The standard resistance coils are made of manganin because :-

(1) Manganin wire is very sensitive to change in temperature

(2) For manganin the temperature coefficient of resistance varies hyperbolically for large range of temperature

(3) For Manganin, the temperature coefficient of resistance is almost zero

(4) None

5. An electric bulb connected to a source having rated voltage consumes more than rated power just after it is switched on, this may be due to :-

(1) As soon as switch is closed a voltage surge occurs across the filament which leads to high power consumption

(2) When filament is at room temperature its resistance is less than its resistance when the bulb is fully illuminated

(3) As soon as switch is closed current through filament increases linearly which leads to high power consumption

(4) because potential difference across filament is less than E.M.F.

6. Electrical conductivity of a typical insulator is about 10^{-22} times that of metal while heat conductivity is about 10^{-3} because :-

(1) In electrical conduction only free e^- take part while in thermal conduction both lattice and free e^- contribute

(2) During electrical conduction no. of free e^- are considerably lesser as compare to the number during thermal conduction

(3) During electrical conduction no. of collision of free e^- are extremely large whereas during thermal conductor no. of collision are small

(4) None

7. Current is passed through a metallic wire, heating it red. When cold water is poured over half of its portion :-

(1) Rest of the portion will become more hot

(2) Rest of the portion will become cold

(3) Whole rod will become cold

(4) None

8. A compartment of a train is lit by 20 identical bulb in series. one of the bulbs fuses and is removed from the chain. The remaining bulbs now light the room. The light in the compartment will :-

(1) Increases
(2) Decreases
(3) Neither increases nor decreases
(4) None

9. Kirchoff's voltage law in a circuit indicates :-

(1) Electric field inside current carrying conductor is non zero
(2) Electric field outside current carrying conductor is always zero
(3) Electrostatic field is conservative field
(4) None of the above

10. A piece of copper and other of germanium are cooled from room temperature to 100 K :-

(1) Conductivity of copper increases and that of germanium decrease
(2) Conductivity of copper decreases and that of germanium increases
(3) Conductivity of copper as well as germanium both decreases
(4) Conductivity of copper as well as germanium both increases

11. A constant potential difference is maintained across a conductor .If the temperature of conductor is increased, drift speed of electron will:-

(1) Increase (2) Decrease
(3) Remain same (4) Can't say

12. On bending a conducting wire its :-

(1) Resistance will increase because drift velocity of electron in bent wire decreases
(2) Resistance will decrease because drift velocity of electron in bent wire increases
(3) Resistance will remain same because drift velocity of elctron in bent wire will remain same
(4) Resistance will remain same but drift velocity of electron in bent wire will decrease

13. The brightness of light bulb, when a heating appliance is switched on will :-

(1) decrease for non-ideal source but remain same for ideal source
(2) Increase for non-ideal source but remain same for ideal source
(3) Remain same for both ideal and non ideal source
(4) Remain same for non-ideal source but decrease for ideal source

14. In which material electric currents develop when an electric field is applied ?

(1) Conductor (2) Wooden piece
(3) Non-conductor (4) Insulator

15. In which substance, positive and negative charges both can move ?

(1) non-electrolytic solution
(2) Electrolytic solution
(3) Both (1) and (2)
(4) Neither (1) nor (2)

16. If we consider a mechanism where, the ends of the cylinder are supplied with fresh charges to make up for any charges neutralised by electrons moving inside the conductor, in that case :-

(1) There will be steady electric field in the conductor
(2) There will be continuous current rather than a current for a short period of time
(3) Both (1) and (2)
(4) Neither (1) nor (2)

17. In $V = \frac{I\rho\ell}{A}$, current per unit area, I/A is called

(1) Resistivity (ρ) (2) Voltage (V)
(3) Current density (J) (4) Resistance (R)

18. When current flows through copper wire, current density J, electric field E and motion of electrons have directions such that :-

(1) J & E in opposite direction
(2) Motion of electrons and E, in opposite direction
(3) J & motion of electrons in same direction
(4) J, E & motion of electrons are in same direction

19. High current in solid conductor can be obtained because :-

(1) drift speed is large

(2) charge on electron is large

(3) electron number density is enormous ~ 10^{29} m^{-3}

(4) None of the above

20. If charges moved without collisions through the conductor, their kinetic energy would also change so that the total energy is :-

(1) Changed (2) Unchanged

(3) Doubled (4) Halved

21. Is it possible that any battery has some constant non-zero value of emf but the potential difference between the plates is zero ?

(1) Not possible

(2) Yes, if another identical battery is joined in supportive and series

(3) Yes, if another identical battery is joined in opposition and series

(4) Yes, possible, if another similar battery is joined in parallel

22. The algebric sum of changes in potential around any closed loop involving resistor and cells in the loop is :-

(1) More than zero (2) Less than zero

(3) Zero (4) Constant

23. The wheat stone bridge and its balance condition provide a practical method for determination of an :-

(1) known resistance

(2) unknown resistance

(3) both (1) and (2)

(4) none of the above

24. Which of the following draws no current from the voltage source being measured ?

(1) Meter bridge (2) Wheat stone bridge

(3) Potentiometer (4) None of these

25. In a potentiometer, the null point is received at 7^{th} wire. If now we have to change the null point at the 9^{th} wire, what should we do ?

(1) Attach resistance in series with cell of secondary circuit

(2) Increase resistance in main circuit

(3) Decrease resistance in main circuit

(4) Decrease applied emf

26. Potentiometer measures the potential difference more accurately than a voltmeter because :-

(1) it has a wire of high resistance

(2) it has a wire of low resistance

(3) it does not draw current from external circuit

(4) it draws a heavy current from external circuit

27. In meter bridge for measurement of resistance the known and the unknown resistances are inter changed. The error so removed is :-

(1) end correction

(2) index error

(3) due to temperature effect

(4) random error

28. Which of the following characteristics of electrons determines the current in a conductor ?

(1) Drift velocity alone

(2) Thermal velocity alone

(3) Both drift velocity and thermal velocity

(4) Neither drift nor thermal velocity

29. Kirchhoff's junction rule is a reflection of:

(1) Conservation of current density vector

(2) Conservation of energy

(3) The fact that the momentum with which a charged particle approaches a junction is unchanged (as a vector) as the charged particle leaves the junction

(4) The fact that there is no accumulation of charges at a junction

30. The direction of the flow of current through electric circuit is :

(1) from low potential to high potential

(2) from high potential to low potential

(3) does not depend upon potential value

(4) current can not flow through circuit

31. The electrical resistance of a conductor depends upon:

(1) Size of conductor

(2) Temperature of conductor

(3) Geometry of conductor

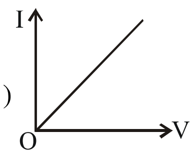
(4) All of these

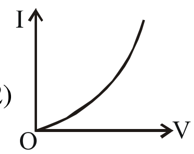
32. For which of the following dependences of drift velocity V_d on electric field E, is Ohm's law obeyed ?

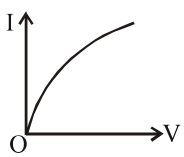
(1) $V_d \propto E$ (2) $V_d \propto E^2$

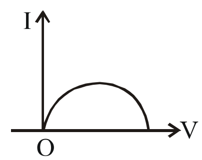
(3) $V_d \propto \sqrt{E}$ (4) $V_d \propto \frac{1}{E}$

33. Which of the following I-V graph represents ohmic conductors ?

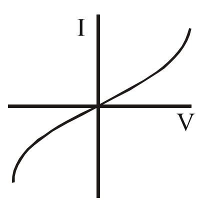
(1)


(2)


(3)


(4)


34. The I-V characteristics shown in figure represents:



(1) Ohmic conductors

(2) Non-ohmic conductors

(3) Insulators

(4) Super conductors

35. Arrange the following materials in increasing order of their resistivity:

Nichrome, Copper, Germanium, Silicon

(1) Copper < Nichrome < Germanium < Silicon

(2) Germanium < Copper < Nichrome < Silicon

(3) Nichrome < Copper < Germanium < Silicon

(4) Silicon < Nichrome < Germanium < Copper

36. With increase in temperature the conductivity of:

(1) Metals increases and semiconductor decreases

(2) Semiconductor increases and metals decreases

(3) Increases in both of them

(4) Decreases in both of them

37. An electric heater is connected to the voltage supply. After few seconds, current get its steady value then its initial current will be :

(1) equal to its steady current

(2) Slightly higher than steady current

(3) Slightly less than its steady current

(4) Zero

38. In the series combination of two or more than two resistances :

(1) The current is same for each resistance

(2) The voltage through each resistance is same

(3) Neither current nor voltage through each resistance is same

(4) Both current and voltage through each resistance is same

39. Principle of wheatstone's bridge is used in a/an :

(1) Galvanometer (2) Potentiometer

(3) Ammeter (4) Voltmeter

40. In a wheat stone bridge if the battery and galvanometer are interchanged the deflection in Galvanometer will :

(1) Change in previous direction

(2) Not change

(3) Change in opposite direction

(4) None of these

41. Point out the right statements about the validity of Kirchoff's junction rule :

(1) Based on conservation of charge

(2) Outgoing currents adds up and are equal to incoming currents at a junction

(3) Bending or reorienting the wire does not change the validity of kirchoff's rule

(4) all of above

42. In parallel combination of n-cells we obtain :

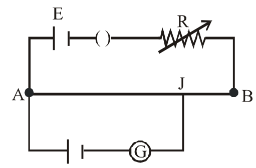
(1) More voltage

(2) More current

(3) Less voltage

(4) Less current

43. AB is a wire of potentiometer. With the increase in the value of resistance R, the shift in the balance point J will be :



(1) towards (B)
(2) towards (A)
(3) remains constant
(4) first towards B then back towards A